

Sebastian Viegas

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EDUCATION

University of Notre Dame

Bachelor of Science in Aerospace Engineering | Minor: Engineering Corporate Practice

GPA: 3.64/4.00

University of Notre Dame Global Gateway Study Abroad

Notre Dame, IN

Expected May 2028

Rome, Italy

May - June 2025

EXPERIENCE

Mach 10 Free-Piston Reflected Shock Tunnel Research Assistant

January 2026 - Present

Corke Lab

Notre Dame, IN

- Developed an original design-condition workflow chaining NASA CEA, PITOT3, and Queensland L1d4, with MATLAB scripts that reconcile differences between the tools and tailor the driver and test gas speed of sound, extending Mach 10 test time from 0.1 ms to 1.8 ms by reducing reflected shock disruption.
- Modeled free-piston shock dynamics and thermochemical nozzle expansion at 5-10 MPa and 1850-3000 K; ran 2D axisymmetric Computational Fluid Dynamics (CFD) in ANSYS Fluent (2M+ cells, as small as 1 micron) to evaluate nozzle exit conditions and boundary layer behavior.
- Ran Finite Element Analysis (FEA) on static and dynamic components to verify a factor of safety of 4 on top of a dynamic load factor of 2.
- Finalized the SolidWorks facility CAD assembly within 600 sq ft of lab space and prepared high-pressure components for DFMA, supplier sourcing, and manufacturing quotes.

Mechanisms Lead & Subsystem Placement Responsible Engineer

September 2024 - Present

IrishSat

Notre Dame, IN

- Lead a student team in design, analysis, and fabrication of satellite mechanisms, including reaction wheels, deployable solar panels, and monopole antennas.
- Architected IrishSat's first fully modular CubeSat and its full-system SolidWorks assembly, validating subsystem fit and tolerances.
- Designed modular reaction wheels, with torque and rotational dynamics calculations verifying rotation on any axis up to 7 deg/s; developed CAM toolpaths with workpiece stabilization to improve accuracy when CNC machining 2 mm thick aluminum wheel components.
- Ran MATLAB thermal simulations of orbital solar radiation on battery housings and built a scale 2U prototype via laser cutting and 3D printing to validate packaging; leading upcoming reaction wheel vibration, thermal, and angular output testing.

UCAH Hypersonic Projectile Flight Competition Project Team Lead

September 2025 - Present

Corke Lab

Notre Dame, IN

- Appointed Team Lead after co-authoring a design report on a low-altitude waverider projectile with L/D above 3; validated shock angles and surface thermal loads on 3D-printed models in a Mach 6 Ludwig wind tunnel.
- Run parametric ANSYS Fluent CFD sweeps (Mach 3-8, -15 to 15 degrees angle of attack, up to 30 km) on 13 million-cell meshes using solution-adaptive refinement on Hessian pressure and shock density gradient, maintaining orthogonal quality above 0.1, to generate lift, drag, and moment coefficients.
- Feed coefficients into a MATLAB three-degree-of-freedom (3DOF) trajectory simulation to maximize range and meet the intended flight trajectory.
- Use CFD heating, pressure, and force data to guide 1D/2D thermal simulations, insulation sizing, FEA, max-Q structural analysis, and internal bulkhead design.

Experimental Liquid Propulsion Team Member

September 2025 - May

2026 Notre Dame Rocketry Team

Notre Dame, IN

- Designed a graphite nozzle insert for liquid propulsion Engine Genesis to withstand 3350 K and 400 psi in the throat and combustion chamber.
- Built a static test stand for hybrid Engine Alpha (waterjet cutting, horizontal bandsaw) and ran dynamic FEA in Fusion 360 at 200 lbf thrust to optimize structural geometry and confirm safety margins.

SKILLS

Software: ANSYS Fluent, MATLAB, SolidWorks, Fusion 360, NASA CEA, Queensland L1d4, Python, Java, Microsoft Suite

Fabrication & Machining: 3D Printing, Waterjet Cutting, Laser Cutting, Machine Shop Tools, GD&T

Aerospace Domains: Hypersonic Aerodynamics, CFD & FEA Simulation, Orbital Mechanisms, Gas Dynamics, Satellite Integration, Liquid Propulsion

Languages: Advanced Portuguese, Intermediate French

Interests: Piano, Hurdling, Music Production